

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Explain how natural selection can lead to antibiotic resistance in bacteria.

[6 marks]

2

Describe an experiment to investigate the effect of light intensity on the rate of photosynthesis.

[6 marks]

3

Explain the difference between mitosis and meiosis and state where each occurs.

[5 marks]

4

Use a Punnett square to predict the offspring ratio from two heterozygous parents.

[4 marks]

5

Describe how the body maintains a constant internal temperature.

[5 marks]

6

Explain the role of the placenta in exchanging substances between mother and fetus.

[4 marks]

7

Describe the structure of a cell membrane and explain how it controls what enters the cell.

[6 marks]

8

Explain the difference between diffusion, osmosis, and active transport, giving one example of each.

[6 marks]
